NATIONAL UNIVERSITY



Fourth Year Syllabus
Department of Soil Science

Four Year B.Sc. Honours Course
Effective from the session : 2013–2014

National University

Subject: Soil Science

Syllabus for Four Year B.Sc. Honours Course

Effective from the session: 2013-2014

Year wise Papers and marks distribution.

FOURTH YEAR

Paper Code	Paper Title	Marks	Credits
243301	Soil Survey and Land Evaluation	100	4
243303	Soil Conservation and Soil Management	100	4
243305	Soil classification and Soils of Bangladesh	100	4
243307	Irrigation and Drainage	100	4
243309	Soil Ecology	100	4
243311	Principles of Agronomy	100	4
243313	Fertilizers and Agricultural Statistics	100	4
243315	Crops of Bangladesh	100	4
243316	Practical	100	4
243318	Viva-Voce	100	4
	Total =	1000	40

Detailed Syllabus

Paper code	243301	Marks: 100	Credits: 4	Class hours: 60 hrs.
Paper title:	Soil Survey and Land Evaluation			

Definition and importance of soil survey. Agricultural and non-agricultural uses of soil survey. Principal types of base materials -their scale, merits and demerits; interpretation and uses of base materials-especially aerial photographs and aerial mosaics; soil survey work-plan.

Type of soil survey: Techniques of Exploratory, Reconnaissance and Detailed soil surveys.

Examination and description of soils in the field; collection of soil samples for laboratory analyses.

Taxonomic unit and mapping unit: Soil maps: units of soil classification and mapping; the soil mapping legend; Stereoscopic examination; plotting of soil boundaries; soil correlation and inspection; soil grouping on the map.

Soil survey report: soil survey interpretation for agricultural and non-agricultural uses.

Land evaluation: Concept of land evaluation; technique and importance of land evaluation, Land use planning, Land capability, Land suitability classification and Crop suitability classification of soil.

Remote sensing - use of remote sensing data for soil survey and land evaluation. Basic principles of remote sensing.

Books Recommended:

1. Soil Survey Manual : USDA
2. Soil Classification with special Reference to the Soils of Bangladesh : Mohammad Sultan Hussain.
3. Soils: Their Survey & Taxonomic Classification : SRDI
4. The Soils of Bangladesh : S. M. Imamul Huq & Jalaluddin Md. Shoaib

Paper code	243303	Marks: 100	Credits: 4	Class hours: 60 hrs.
Paper title:	Soil Conservation and Soil Management			

Soil as a basic & irreplaceable resource: Population versus resource base of the world.

Soil Degradation: Types and Processes of soil degradation, Factors of soil degradation.

Soil Erosion: Classification of Soil erosion, Harmful effects of soil erosion, factors affecting water & wind erosion. Impact of erosion in Bangladesh.

Soil conservation: past, present and future approaches of soil conservation, principles of soil conservation; planning soil conservation for a given area; Problems of implementing soil conservation practices in the field; socio-economic aspects of soil conservation.

Sustainable land use: defining sustainability; sustainable land use system and soil resilience; assessment of soil sustainability; soil quality concept; physical, chemical and biological indicators of soil quality.

Objectives and principles of soil management: Preparation of seedbed; maintenance of organic matter in soil; Nitrogen economy of soils. Organic manures- green manure, farm manure, compost, preparation of different types of manures and composts; microbiology of manures and composts; Importance of manures and composts in improving soil properties and crop yield.

Origin and nature of problem soils; formation, properties. amelioration and management of various problematic soils; saline, alkali and acid soils, waterlogged soils, hilly soils, forest soils. sandy soils. Organic soils and their utilization. Concept of dry farming and organic farming.

Books Recommended:

1. Soil Conversation and Sustainable Landuse : T. H Khan
2. Advance in Soil Science, Vol II: Soil Degradation : R. Lal and B.A. Stewart
3. Methods for Assessment of Soil Degradation : W. H. Blum, C. Valentine and B. A. Stewart (Ed.)
4. Soil Erosion and Conservation : R. P. C. Morgan
5. Soil Management of India : H. R.. Arakari, G. V. Chalan, P. Satyanarayan, R. L. Donahue
6. Problem Soils of Bangladesh : BARI
7. Manual of Soil & Water Conservation Practices : Gurumel Singh. C.Venkatarmanan

Paper code	243305	**Marks: 100**	**Credits: 4**	**Class hours: 60 hrs.**
Paper title:	**Soil Classification and Soils of Bangladesh**			

Geographical distribution of major soils on the Earth's surface; Zonality concept of soils; Great groups of soils; Soil Taxonomy: Study of the orders and sub-orders of soil taxonomy; Nomenclature in Soil Taxonomy.

Environmental conditions and pedogenic factors in Bangladesh; Dominant soil forming processes in Bangladesh; Inundated land types; General conditions of soils in Bangladesh; Physiographic units in Bangladesh. Land Capability & Crop suitability Classification of the soils of Bangladesh.

Major soils of Bangladesh;7 Soil tracts; General soil types of Bangladesh: their distribution and properties; Classification of the soils of Bangladesh on the basis of Soil Taxonomy and FAO-UNESCO Legend: Agro-ecological Zones of Bangladesh: Lower level classification of Bangladesh soils-soil series and families.

Books Recommended:

1. Soil Classification with Reference to the Soils of Bangladesh : M.S. Hussain
2. Soils of Bangladesh: their Genesis, Classification and Use Potential : M.M. Hassan
3. Soils of Bangladesh : M.R. Rahman
4. The Geography of the Soils of Bangladesh : H. Brammer
5. SOILS: Their survey & Taxonomic Classification : SRDI

Paper code	243307	Marks: 100	Credits: 4	Class hours: 60 hrs.
Paper title:	Irrigation and Drainage			

The field water regime**.** field water balance. Infiltration - factors affecting infiltration; profile moisture distribution during infiltration; methods of measuring infiltration rate. Redistribution of water in soil profile. Field capacity concept. Evaporation - factors affecting evaporation; evaporation from bare soil surface; reduction of evaporation from bare soil. Evapotranspiration, Consumptive use and water use efficiency. Methods of measuring evapo-transpiration.

Irrigation: sources and quality of irrigation water; selection of land for irrigation; factors affecting irrigation needs; amount and frequency of irrigation, soil and plant-water parameters for determining irrigation needs; different irrigation systems - their suitability for different soils and crops, irrigation efficiency, effect of irrigation on soil properties, irrigation needs of major crops of Bangladesh; irrigation systems and irrigation projects in Bangladesh.

Drainage: needs and benefits of drainage; types of drainage - their advantages and disadvantages; drainage of coastal plains and swamplands. Factors affecting drainage, drainage equations.

Books Recommended:

1. Manual on Irrigation Agronomy **:** R.D. Misra.
2. Irrigation **:** S. R. Sahasrabudhe.
3. Principles of Irrigation **:** A.M.Michael
4. Irrigation **:** Hudson
5. Principles and Practices of Irrigation Engineering. **:** S. K. Sharma

Paper code	243309	Marks: 100	Credits: 4	Class hours: 60 hrs.
Paper title:	Soil Ecology			

Ecology and its domain. Communities and Ecosystems: species abundance and diversity; primary production and energy flow; nutrient cycling and retention; succession and stability. Soil ecosystem and ecological community.

Organisms: match between organisms and their environments; conditions, resources, life and death in Unitary and Modular organisms, dispersal, dispersion and migration in space and time.

Interactions among species- competition; exploitation, predation, herbivory parasitism and disease, symbiosis and mutualism.

Soil ecology: main features of soil ecology; soil ecological effects on plants, microbial biomass and biodiversity; manipulation of soil ecology; soil biotechnology.

Sources and sinks of greenhouse gases; response of soils to global climatic change; gaseous exchange; fluxes and their impact on ecosystem .

Agro-ecosystems: Soil and agriculture; Effects of agriculture on environment.

Forest ecosystems: Biomes; evolution and diversity in biomes, dominant forest vegetation on earth; species diversity; plant forms and adaptation; ecology of the forests in Bangladesh.

Books Recommended:

1. Soil Ecology : Ken Killham
2. Ecology : M. Begon.
3. Fresh Water Ecology : Michel Jeffries.
4. Environmental Management of Fresh Water Ecology : Trivedi.
5. Ecology: Concept and Applications : Manuel. C. Molles Jr.

Paper code	243311	**Marks: 100**	**Credits: 4**	**Class hours: 60 hrs.**
Paper title:	**Principles of Agronomy**			

Origin and importance of agriculture; agronomy and duties of an agronomist; location, area, physiography, major rivers, climate, agricultural seasons and crops of Bangladesh.

Weather and climate - their influence on vegetation and crop production; agronomic classification of plants.

Soil tillage – purposes and methods of tillage practices; tillage equipment and their classification; quality of seedbed and root-bed.

Seeds: definition and importance, seed testing and seed treatments; seed germination, healthy and crippled seedlings; quality of good seeds and factors affecting quality; certified seeds; seed law and seed association.

Weeds: classification of weeds; harmful and beneficial effects of weeds; crop-weed association; weed management and weed control; common weeds of Bangladesh.

Cropping pattern - orientation with respect to ecological environment in Bangladesh.

Horticulture and Floriculture in Bangladesh. Peri-urban agriculture and its importance. General survey of agricultural data in relation to the production of different crops in Bangladesh

Farm layout, farming and farm budgeting; **crop calendar**- principles of its preparation and its utility.

Books Recommended:

1. Fundamentals of Agronomy : C Gopal Chandra. De,-2011
2. Principles and Practices of Agronomy : S. S. Sing
3. Principles of Weed Science : V.S. Rao
4. Agricultural chimatology : J.R. Kakde

Paper code	243313	**Marks: 100**	**Credits: 4**	**Class hours: 60 hrs.**
Paper title:	**Fertilizers and Agricultural Statistics**			

Fertilizer materials: Sources of fertilizer materials; manufacture of major fertilizers; fertilizer grades and fertilizer ratios; fertilizer law; organic manures and bio-fertilizers.

Fertilizer movement in soil; fate of added nutrient in soil; residual effects of fertilizers; salt index; balanced fertilization; impact of fertilizer application on soil quality and environment.

Pesticides and Fungicides: definition, classification, use and impact on soil and water ecosystem; biodegradation of pesticides and fungicides;

Statistics- its meaning, scope and uses: importance of statistics in agriculture and environment.

Data collection; classification, tabulation and graphical representation, normal distribution, frequency and frequency distribution.

Central tendency**;** measurement of central tendency; mode, median, quartile, rank and range.

Dispersion: measures of dispersion- skewness, kurtosis, variance, standard deviation, standard error. Confidence limit and coefficient of variation .

Population and sampling: types of sampling, size of sampling, role of normal distribution in statistics;sample mean, chi-square distribution, F-distribution, students t-distribution, normal test, t-test and X2-test.

Probability and hypothesis testing**:** probability rules, null hypothesis, and choice of test.

Books Recommended:

1. Soil Fertility and fertilizer : Samuel L. Tisdale, Werner L. Nelson and James D. Beaton.
2. Simples Lessons from Biometry : Zaman & Zaman, BRRI, Gazipur.
3. Agricultural Statistics : Gomaz & Gomaz.
4. সামাজিক পরিসংখ্যান পরিচিতি : মো: আবদুর রাজ্জাক
5. Fertilizer recommendation Guide-2012 : BARC

Paper Code	243315	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Crops of Bangladesh**			

Classification of crops.

Agronomy of arable crops of Bangladesh: Origion, climate, soil requirements, cultural practices, fertilization, harvesting and storing of the main crops of Bangladesh.

Cereal crops of Bangladesh: Rice, Wheat and Maize. Brief description about other minor cereal crops of Bangladesh.

Fibre crops of Bangladesh: Jute and Cotton. Brief description about other minor fibre crops.

Other important crops of Bangladesh: Sugarcane, Potato, Tobacco, Pulse and Oilseeds.

Agronomy of perennial crops: Tea, Rubber, Jackfruit, Mango and coconut.

Cropping systems practised in Bangladesh: Cropping Sesons, Mono-cropping, Mixed cropping, Multiple cropping, Crop rotation and Cropping intensity.

Book Recommended :

1. কৃষি প্রযুক্তি হাত বই - BARI-2012
2. বাংলাদেশের খাদ্য শস্য ও অর্থকারী ফসল - বাংলা একাডেমী
3. বাংলাদেশের ফল চাষ - বাংলা একাডেমী
4. ধানচাষ-২০১২ - BARI

Paper code	243316	**Marks: 100**	**Credits: 4**	**Class hours: 60 hrs.**
Paper title:	**Practical**			

1. Soil survey of a given area: study of base materials**;** stereoscope study and interpretation of aerial photograph; study of soil in the field, demarcation of soil boundary; preparation of soil map, soil survey report and recommendations.

2. Measurement of physical, chemical and biological indicators for assessing the quality of soil of a given area. Preparation of report and recommendation.
3. Visit to the polluted areas and collection of waste, effluent, soil and plant samples; determination of heavy metals in waste, sewage sludge, soil and plant sample, Determination of water quality - DO, BOD, COD, Cl-, HCO3-, CO32-, NO3-, NH+4, and SO42-.Report preparation and recommendation.

Books Recommended:

1. A Hand book on Analysis of SOIL, PLANT & WATER – S. M. Imamul Huq & Md. Didar ul A.am
2. Methods in Environmental Analysis – Soil, Air & Water – P. K. Gupta.

Paper code	243318	**Marks: 100**	**Credits: 4**	
Paper title:	**Viva-Voce**			